Separable Equations

Section 1: Separation Form and Integrals

1 A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is separable if it can be factored into:

a. $\frac{dy}{dx} = g(x)h(y)$

b. $\frac{dy}{dx} = g(x) + h(y)$

c. $\frac{dy}{dx} = g(x/y)$

d. $\frac{dy}{dx} = x^n + y^n$

2 Separate variables for $\frac{dy}{dx} = 3x^2y^2$.

a. $\frac{1}{y^2}dy = 3x^2dx$

b. $y^2dy = 3x^2dx$

c. $\frac{1}{3x^2}dy = y^2dx$

d. $dy = 3x^2y^2dx$

3 Solve the general equation $\frac{1}{y^2}dy = 3x^2dx$.

a. $y = -\frac{1}{x^3 + C}$

b. $y = \frac{1}{x^3 + C}$

c. $y = -x^3 + C$

d. $y = \sqrt{x^3 + C}$

Section 2: Initial Value Problems (IVPs)

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Solve the IVP: $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = -3$.

a. $y = -\sqrt{x^2 + 9}$

b. $y = \sqrt{x^2 + 9}$

c. $y = -\sqrt{x^2 - 9}$

d. $y = -x - 3$

5 Solve $\frac{dy}{dx} = 2y$ with $y(0) = 7$.

a. $y = 7e^{2x}$

b. $y = 7e^{-2x}$

c. $y = 2e^{7x}$

d. $y = 7 + 2x$

6 What is a singular solution of a differential equation?

a. A solution that cannot be obtained from the general family for any value of the arbitrary constant C

b. A solution that blows up to infinity everywhere

c. A solution where the derivative is undefined

d. A solution with no arbitrary constants

Section 3: Exponential and Logistic Growth Models

7 What is the differential equation for logistic population growth with carrying capacity K and growth rate r?

a. $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$

b. $\frac{dP}{dt} = r(K - P)$

c. $\frac{dP}{dt} = rP^2$

d. $\frac{dP}{dt} = \frac{rP}{K}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 At what population size does the logistic growth rate $\frac{dP}{dt}$ reach its maximum?

a. $P = \frac{K}{2}$

b. $P = K$

c. $P = 0$

d. $P = \frac{K}{4}$

9 What is $\lim_{t \to \infty} P(t)$ in a logistic model if $P(0) > 0$?

a. K

b. ∞

c. 0

d. rK

Section 4: Newton's Law of Cooling

10 Newton's Law of Cooling states that the rate of change of temperature T is proportional to the difference between T and the ambient temperature T_m:

a. $\frac{dT}{dt} = -k(T - T_m)$ with $k > 0$

b. $\frac{dT}{dt} = -kT$

c. $\frac{dT}{dt} = k(T + T_m)$

d. $\frac{dT}{dt} = -k(T - T_m)^2$

11 A metal bar at 100°C is placed in a room at 20°C. If $T(t) = 20 + 80e^{-kt}$ and cools to 60°C after 10 minutes, find e^{-10k}.

a. $\frac{1}{2}$

b. $\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 Using $e^{-10k} = \frac{1}{2}$ from above, what is the temperature after 20 minutes?

a. $40^\circ C$

b. $30^\circ C$

c. $50^\circ C$

d. $25^\circ C$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647